Basic Local Alignment Search Tool

BLAST

Making sense of information

Suppose your sequencing experiment generates:

```
>mystery
TGATGATGTTGACCAAAGTTTGATTATCGCTGCTAGAAACATAGTAAGAAGAGC
AGCAGTGTCAGCAGAC CCATTAGCATCTCTCTTGGAAATGTGCCACAGCACAC
AGATTGGAGGTGTGAAGATGGTGGACATCCTTAGACAGAATCCAACTGAGGAAC
AAGCCGTGGACATATGCAAGGCAGCAATAGGGTTGAGGATCAGCTCATC
```

1. Have you discovered something new? Has this been published?
2. Is this sequence similar to a known sequence?
3. Is **any part** of this sequence similar to **any other part** of known sequences?

Searching EVERYTHING

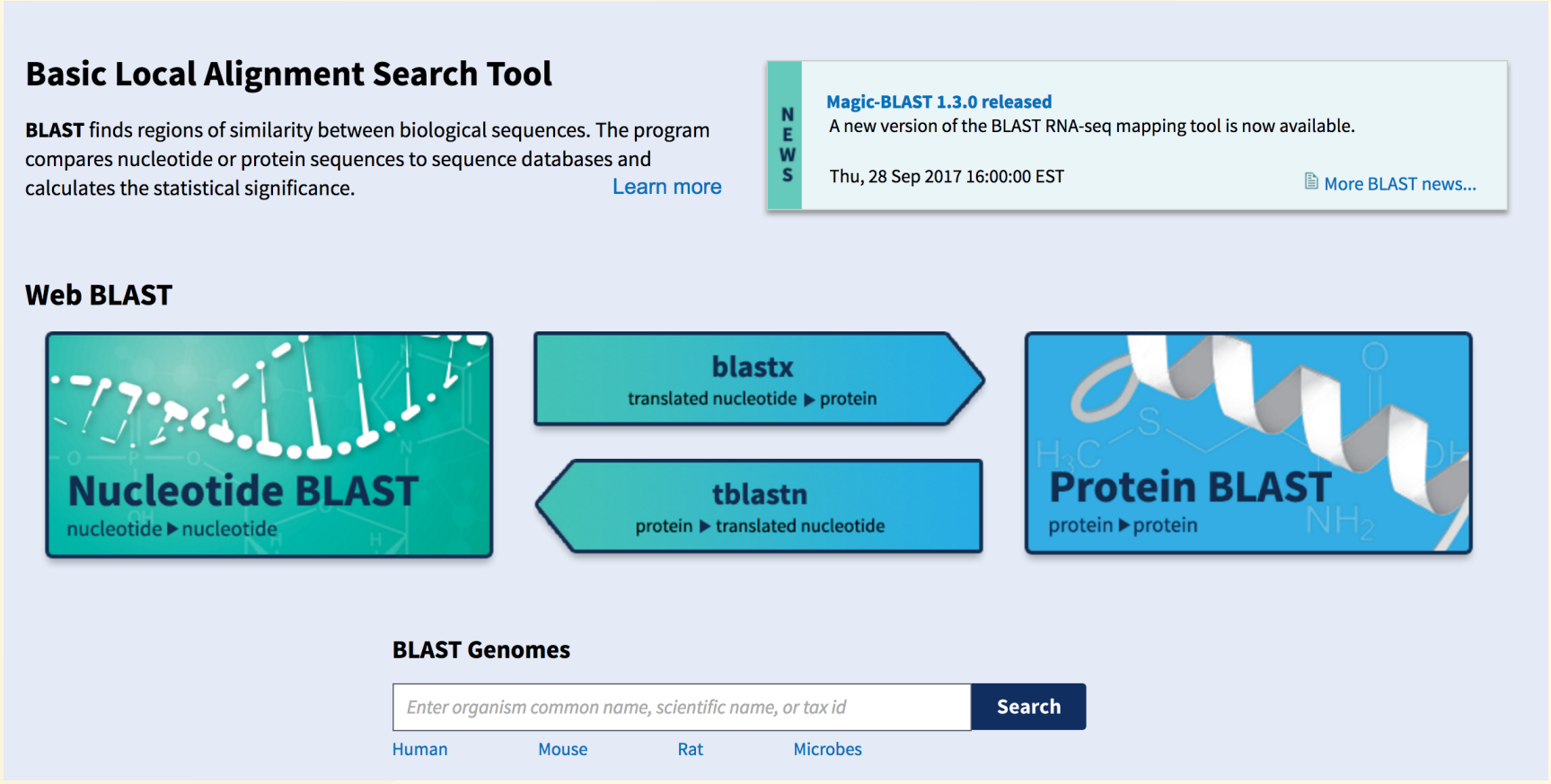
The need to search all known information is a basic need of science.

BLAST (Basic Local Alignment Search Tool) is a methodology to search **few and short sequences** against a collection (potentially millions) of sequences.

It is the most popular, probably the most commonly **used** (and **misused**) bioinformatics tool.

Cited over 60 thousands times. Among the most cited scientific publications ever.

BLAST web interface




```
wget http://data.biostarhandbook.com/fasta/mystery1.fa
```

BLAST Applications

Standalone and API BLAST

Download BLAST
Get BLAST databases and executables

Use BLAST API
Call BLAST from your application

Use BLAST in the cloud
Start an instance at a cloud provider

Specialized searches

SmartBLAST
Find proteins highly similar to your query

Primer-BLAST
Design primers specific to your PCR template

Global Align
Compare two sequences across their entire span (Needleman-Wunsch)

CD-search
Find conserved domains in your sequence

GEO
Find matches to gene expression profiles

IgBLAST
Search immunoglobulins and T cell receptor sequences

VecScreen
Search sequences for vector contamination

CDART
Find sequences with similar conserved domain architecture

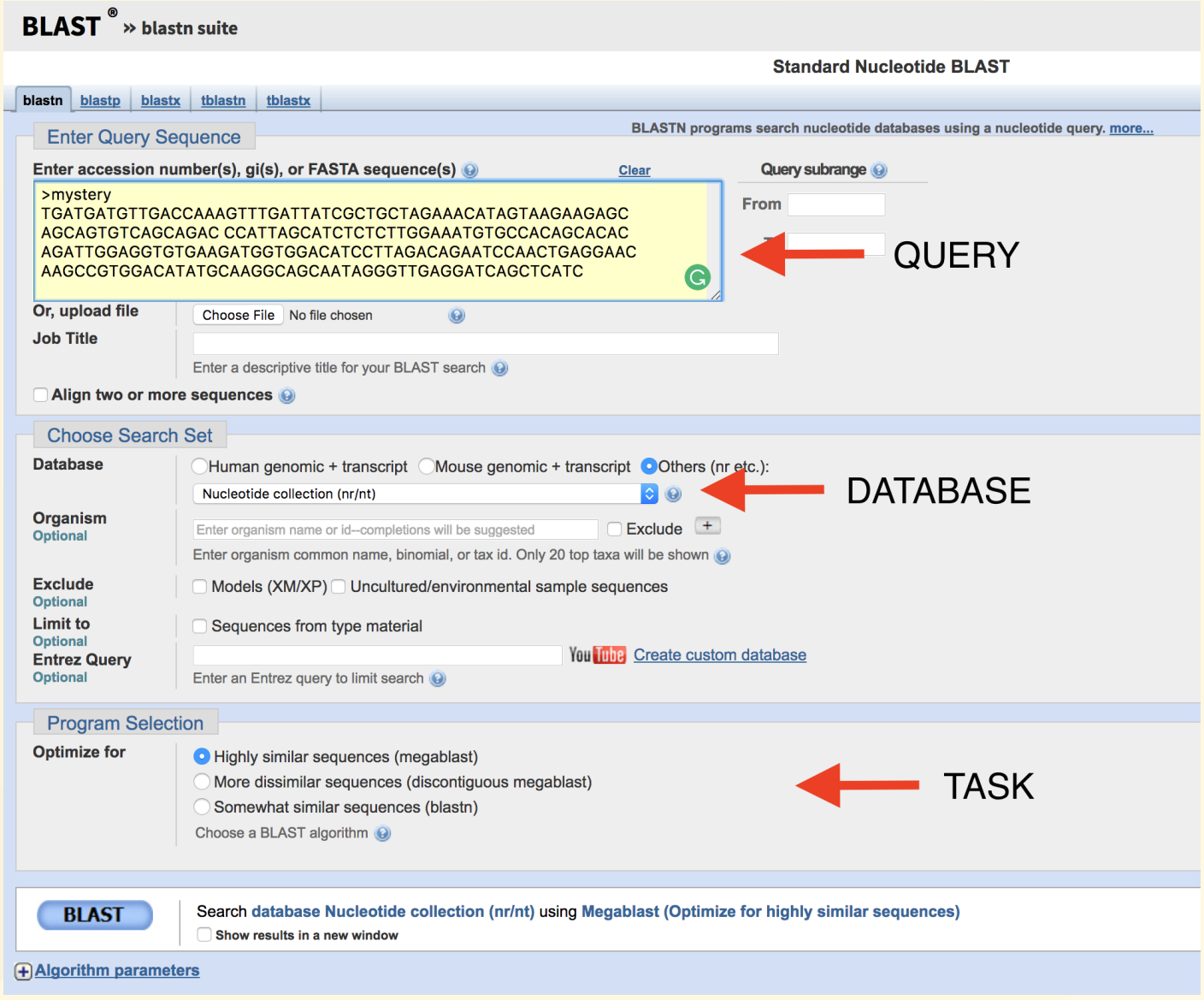
Targeted Loci
Search markers for phylogenetic analysis

Multiple Alignment
Align sequences using domain and protein constraints

BioAssay
Search protein or nucleotide targets in PubChem BioAssay

MOLE-BLAST
Establish taxonomy for uncultured or enviromental sequences

Web interface to BLAST



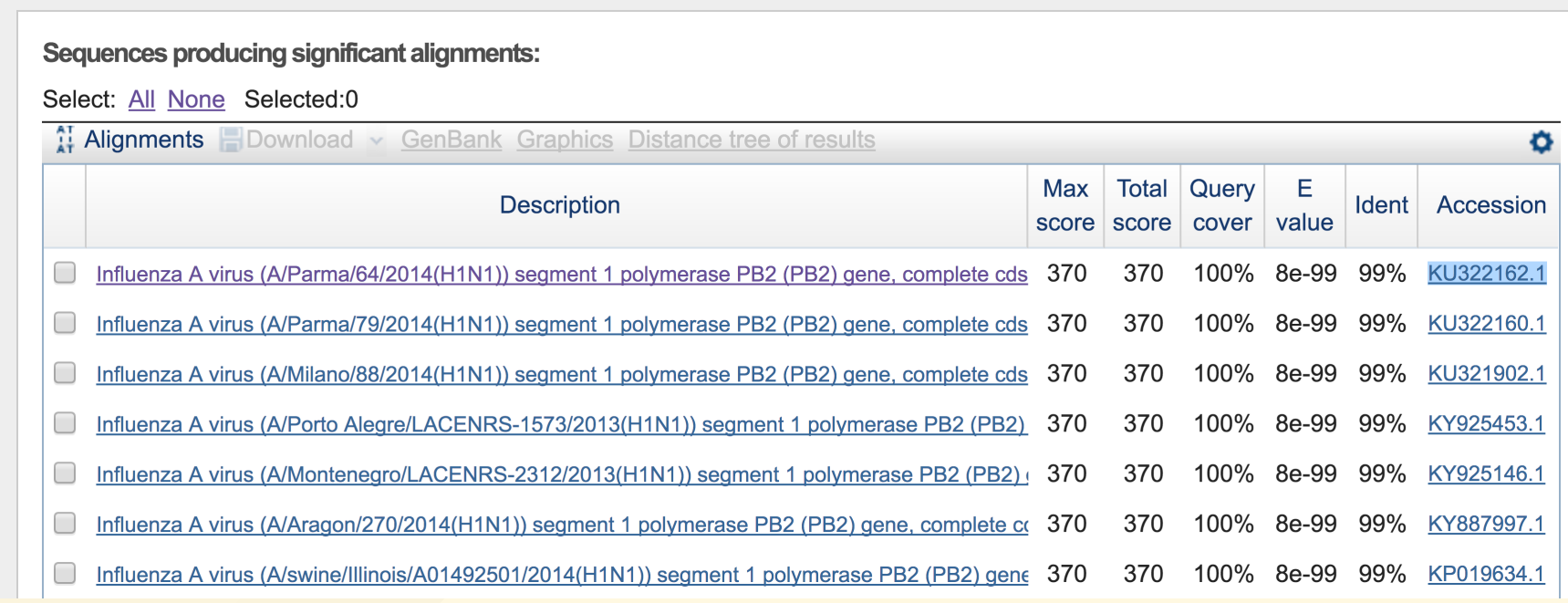
BLAST Results

Get the sequence

```
wget http://data.biostarhandbook.com/fasta/mystery1.fa
wget http://data.biostarhandbook.com/fasta/mystery2.fa
```

Run web BLAST

- Use `mystery1.fa`



Sequences producing significant alignments:

Select: All None Selected:0

Alignments Download GenBank Graphics Distance tree of results

Description	Max score	Total score	Query cover	E value	Ident	Accession
Influenza A virus (A/Parma/64/2014(H1N1)) segment 1 polymerase PB2 (PB2) gene, complete cds	370	370	100%	8e-99	99%	KU322162.1
Influenza A virus (A/Parma/79/2014(H1N1)) segment 1 polymerase PB2 (PB2) gene, complete cds	370	370	100%	8e-99	99%	KU322160.1
Influenza A virus (A/Milano/88/2014(H1N1)) segment 1 polymerase PB2 (PB2) gene, complete cds	370	370	100%	8e-99	99%	KU321902.1
Influenza A virus (A/Porto Alegre/LACENRS-1573/2013(H1N1)) segment 1 polymerase PB2 (PB2)	370	370	100%	8e-99	99%	KY925453.1
Influenza A virus (A/Montenegro/LACENRS-2312/2013(H1N1)) segment 1 polymerase PB2 (PB2)	370	370	100%	8e-99	99%	KY925146.1
Influenza A virus (A/Aragon/270/2014(H1N1)) segment 1 polymerase PB2 (PB2) gene, complete c	370	370	100%	8e-99	99%	KY887997.1
Influenza A virus (A/swine/Illinois/A01492501/2014(H1N1)) segment 1 polymerase PB2 (PB2) gene	370	370	100%	8e-99	99%	KP019634.1

We get a hit for KU322162.fa

(and many others)

Verify the results

BLAST found us candidates such as `KU321902.fa`

Get the data and align with local alignment:

```
efetch -db nuccore -id KU322162 -format fasta > KU322162.f
```

then to produce a local alignment run:

```
local-align.sh mystery1.fa KU321902.fa
```

what would a global alignment look like:

```
global-align.sh mystery1.fa KU321902.fa
```

```
mystery       1 TGATGATGTTGACCAAAGTTGGATTATCGCTGCTAGAAACATA
                |||||||||||||||||||.||||||||||||||||||||||
KU321902.1  756 TGATGATGTTGACCAAAGTTTGATTATCGCTGCTAGAAACATA

mystery1     51 GAGCAGCAGTGTCAGCAGACCCA-TAGCATCTCTCTTGGAAAT
                ||||||||||||||||||||||| ||||||||||||||||||||
KU321902.1  806 GAGCAGCAGTGTCAGCAGACCCATTAGCATCTCTCTTGGAAAT

mystery1    100 AGCACACAGATTGGAGGTGTGAAGATGGTGGACATCCTTAGAC
                ||||||||||||||||||||||||||||||||||||||||||||
KU321902.1  856 AGCACACAGATTGGAGGTGTGAAGATGGTGGACATCCTTAGAC

mystery1    150 AACTGACGAACAAGCCGTGGACATATGCAAGGCAGCAATAGGG
                ||||||.||||||||||||||||||||||||||||||||||||
KU321902.1  906 AACTGAGGAACAAGCCGTGGACATATGCAAGGCAGCAATAGGG

mystery1    200 TCAGCTCATC    209
                ||||||||||
KU321902.1  956 TCAGCTCATC    965
```

The great thing about BLAST

You search everything.

You may find all kinds of interesting stories that can be published.

It is fast and efficient and can be tuned via parameter settings.

The NOT so great thing about BLAST

You search everything.

You may find large number of fake stories that you may not be able to separate from true stories.

It can be very sensitive to parameter settings.

Most scientists don't understand the limitations of the method.

The difference between the file

`mystery2.fa` is the same as `mystery1.fa` only that it has three more sequence lines

```
diff -u mystery1.fa mystery2.fa

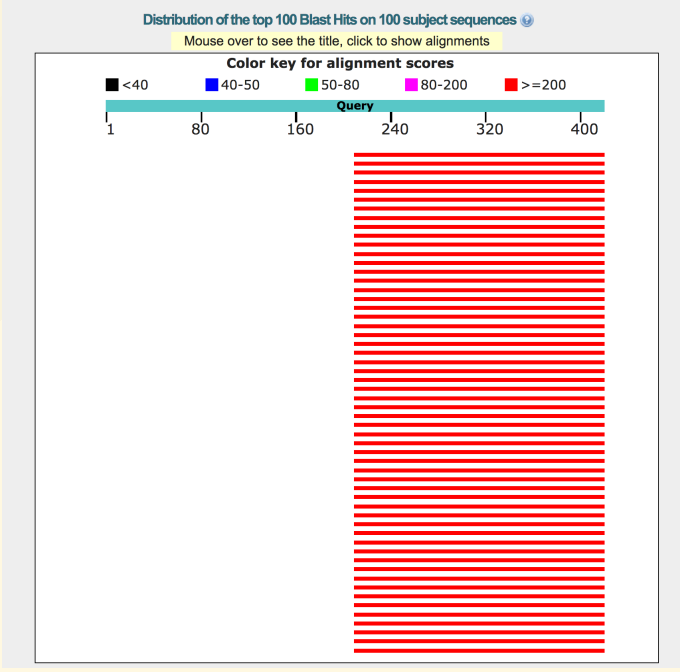


--- mystery1.fa 2017-10-03 12:43:57.000000000 -0400
+++ mystery2.fa 2017-10-04 10:21:12.000000000 -0400
@@ -1,4 +1,7 @@
->mystery1
+>mystery2
 TGATGATGTTGACCAAAGTTGGATTATCGCTGCTAGAAACATAGTAAGAAGAGCAGC
 CCATAGCATCTCTCTTGGAAATGTGCCACAGCACACAGATTGGAGGTGTGAAGATGG
 GACAGAATCCAACTGACGAACAAGCCGTGGACATATGCAAGGCAGCAATAGGGTTGA
+GATTAATAATTTTCCTCTCATTGAAATTTATATCGGAATTTAAATTGAAATTGTTAC
+GGTTTGTTTCAGAGCCATATCACCAAGATAGAGAACAACCTAGGTCTCCGGAGGGGG
+GTGCTCAGTTGAAAATCCCTTGTCAACATCTAGGCCTTATCACATCACAAGTTCCGC
```

This is how BLAST can be misused

Even though the sequence is half Flu + half Ebola the default BLAST will not find Influenza anymore.

It generates hundreds of hits of Ebola Virus, triggers various cutoffs and stops reporting.



Command line BLAST

If all you need is to search a few sequences, web interface to blast is sufficient.

When do you need a command line interface:

- For a more systematic search
- To automate and repeat searches
- To overcome web based limitations

Running BLAST at the command line

To run BLAST you need a database and a query.

A BLAST database is a FASTA file reformatted to work with BLAST.

You can make your own BLAST databases, or you can download pre-built databases.

Get the query

Get `mystery3.fa`. We know that it is some sort of 16S marker gene.

```
wget http://data.biostarhandbook.com/fasta/mystery3.fa
```

You can download prebuilt databases

NCBI provides some standard databases:

```
update_blastdb.pl  --showall | head

16SMicrobial
cdd_delta
env_nr
...
```

Get the

Get the prepackaged databases for `16S` genes:

```
update_blastdb.pl --decompress 16SMicrobial
```

Run `blastn` with your query:

```
blastn -query mystery3.fa -db 16SMicrobial
```

Most often we place the results in a file:

```
blastn -query mystery3.fa -db 16SMicrobial > results.txt
```

Understanding the BLAST output

```
>NR_114736.1 Raoultella ornithinolytica strain JCM6096 16S
partial sequence
Length=1516

 Score = 475 bits (257),  Expect = 1e-133
 Identities = 490/589 (83%), Gaps = 78/589 (13%)
 Strand=Plus/Minus

Query  1    TTTAACCTTGCGGCCGTACTCCCCAGGCGGTCGATTTAACGCGTTA
            |||||||||||||||||||||||||||||||| ||||||||||
Sbjct  882  TTTAACCTTGCGGCCGTACTCCCCAGGCGGTCGACTTAACGCGTTA

Query  61   GCCTCAAGGGCACAACCTCCAAATCGACATCGTTTACAGCGTGGAC
             ||||||||| |||||||||| ||||||||||||||||||||||||
Sbjct  822  TCCTCAAGGGAACAACCTCCAAGTCGACATCGTTTACAGCGTGGAC
```

BLAST offers different output formats

There are simpler to interpret outputs

```
blastn -query mystery3.fa -db 16SMicrobial -outfmt 7 | hea

# BLASTN 2.6.0+
# Query: mystery3
# Database: 16SMicrobial
# Fields: query acc.ver, subject acc.ver, % identity, alig
# 500 hits found
mystery3        NR_025635.1     85.932  590     3       65
mystery3        NR_117686.1     85.593  590     5       65
mystery3        NR_117685.1     85.593  590     5       65
mystery3        NR_117684.1     85.593  590     5       65
mystery3        NR_117682.1     85.593  590     5       65
```

This is output format 7

Using BLAST effectively means knowing how to select which information gets displayed.

BLAST help is difficult to read

```
blastn -help
```

prints pages, starting with:

```
USAGE
blastn [-h] [-help] [-import_search_strategy filename]
[-export_search_strategy filename] [-task task_name] [-db
[-dbsize num_letters] [-gilist filename] [-seqidlist filen
[-negative_gilist filename] [-entrez_query entrez_query]
[-db_soft_mask filtering_algorithm] [-db_hard_mask filteri
[-subject subject_input_file] [-subject_loc range] [-query
[-out output_file] [-evalue evalue] [-word_size int_value]
[-gapopen open_penalty] [-gapextend extend_penalty]
[-perc_identity float_value] [-qcov_hsp_perc float_value]
```

It takes some perseverance to wade through that.

Check the formatting in the help

```
*** Formatting options
-outfmt <String>
  alignment view options:
...

Options 6, 7, 10 and 17 can be additionally configured to
a custom format specified by space delimited format speci
The supported format specifiers for options 6, 7 and 10 a
...
   qseqid means Query Seq-id
      qgi means Query GI
     qacc means Query accesion
  qaccver means Query accesion.version
     qlen means Query sequence length
```

This is perhaps the most important feature. You will have to format the output to contain what you need.

Printing only the fields you want

```
blastn -query mystery3.fa -db 16SMicrobial
      -outfmt "6 qseqid qlen sacc slen pident" | head -5
```

produces a listing where the last column is `pident` (percent identity):

```
mystery3        610     NR_025635       1412    85.932
mystery3        610     NR_117686       1530    85.593
mystery3        610     NR_117685       1530    85.593
mystery3        610     NR_117684       1530    85.593
mystery3        610     NR_117682       1524    85.593
```

You can sort this table by various columns of interest.

How does BLAST work?

A database called **target** (**subject**) contains one or more sequences (could be even millions!)

BLAST searches the **target** sequences with a **query** sequence.

BLAST produces a list of local alignments where the **query** is similar to one or more **target** sequences.

BLAST concepts

Search may take place in nucleotide and/or protein space

Searches implement additional optimizations called tasks: **megablast**, **blastn-short**...

Searches rely on scoring matrices

Searches may be customized with many other parameters. It has many many subtle functions that most users never need.

Knowing BLAST can be a “standalone job”. There are (outdated) books written on just BLAST.

How does BLAST work?

Finds very short exact matches (seeds)

Extends the short exact matches to longer regions.

Performs optimal alignment on the extended regions.

Then applies a number of filtering on the results to reduce the results. This is what misleading reports may be caused by.

We have to reduce data - since there could be suprious and hits by random chance. Care must be taken to not remove essential information either.

Use blast+

Two versions for the program:

1. **blast+**, after 2013, uses programs such as **makeblastdb**, **blastn**, **blastp** and has search tasks such as *megablast*
2. **blast**, before 2013, uses programs such as **formatdb**, **blastall**, **megablast** and has search strategies such as *blastn*, *blastp*

There is still quite a bit of documentation that refers to the old blast.

Running blast

You need a **subject/target** space (database): Where are you searching.

You need a **query** sequence: What are you searching for?

You need a **search strategy**: How are you looking for the query?

Blast search space

BLAST can search nucleotide and protein space.

It can also translate nucleotides into proteins both for the query and the database.

When it translates a sequences it does six times more work per translation (three reading frames and two strands).

If translates both then it needs to do 6x6 = 36 times more work.

Blast program naming

It would be easier to remeber which `blast` program does what the name it encoded the query/target types.

Query	Target	Space	Ideal Name	Real Name
nucl	nucl	nucl	blastNN	**blastn**
pept	pept	pept	blastPP	**blastp**
nucl	pept	pept	blastNP	**blastx**
pept	nucl	pept	blastPN	**tblastn**
nucl	nucl	pept	blastNNP	**tblastx**